Resume
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The year 2023 marks a turning point for researchers and learners. Generative AI has changed my life. It's just like how the X-ray was invented in 1895, which opened a door to many discoveries later in the 20th century. Now, people who aren't smart can also do great work. where a discovery mindset and motivation for innovation are prioritized over technical skills more than ever before. I have been following the development in the GenAI field very closely. My research will use AI in every step.

Research/Work

- **Research Assistant for Precision Swine Production - University of Missouri 07/2024 - 12/2024**
- Developed solutions to predicate estrous time in swine: configured MCU, and LIDAR/IR cameras with codes to become an inspection robot; trained in models for imaging segmentation and transformer neural network. Filled an NSF fellowship proposal. a two-page extended abstract proposed the ground true uncertainty issue might be addressed by using microneedle patch for continuous estradiol monitoring
- **Bioinformatics Research Assistant - Florida Atlantic University/Texas Tech University, 03/2023 - 06/2023, 04/2024 - 06/2024**
- Performed mutation and transposable element between samples from whole exome/genome data,, using GATK/BWA/Samtools/Bcftools/Pysam/Snpeff/customized python code, short term contractor working with the same team.
- **Research Data Assistant - IU School of Medicine, 06/2021 - 07/2022**
- Analyzed over 50 various RNA-seq projects at sequencing core, included: processing raw RNA sequencing data, mapping, sample outlier detection, batch effect removing, various DGE analysis, pathway analysis, splicing analysis, troubleshooting
- Generated html report with the newest visualization tools/plots for the users
- Processed raw WES data, SNP, INDEL analysis
- Developed a GUI database
- Provided consultation to researchers regarding the analysis result
- Mentored undergraduate and master students

Education

- **Arizona State University, 01/2025 – present, M.S. Electrical Engineering, online**
  Took VLSI design, Digital Signal Processing (DSP) courses

- **San Diego State University, 01/2025 – present, M.S. Medical Physics, onsite**
  Courses in medical physics, clinical lab in diagnostics imaging and therapy, CAMPEP program. Shadowed in a MEMS lab with research using electrochemistry and graphene to monitor glucose level in real time. Research in blockchain data systems for agriculture and medical imaging applications. Teaching Associate for electricity lab. Selected for two fellowships in research. Expected all courses completed in Fall 2025.

- **University of Missouri Columbia, 08/2024-12/2024, master student**
  Took courses in VHDL digital logic devices, CMOS, Virtuoso, Vivado

- **Purdue University, 08/2023-05/2024, non-degree student taking medical physics courses,**
  Took courses about how imaging begging generated through X-ray, CT, PET, MR, or ultrasound; conducted MR research that isolated non-water proton signals, led in a poster presentation at the school conference; developed TG51 online calculator for radiation dosage calculation led to oral presenting in *regional* conference

2017-2021

- **Indiana University – Purdue University Indianapolis (IUPUI), B.S. in Biomedical Informatics**

150 credits in life science, statistics, computer science, bioinformatics, Internship to identify potential pathways and markers related to mitochondria in RGC cells from single cell sequencing data.

- **Saint Louis Community College, University of Nebraska Omaha, Ivy Tech Community College**
  NSF funded 16 credits hands-on training in biotechnology, Indiana General Education Transfer Core Certificate

Projects

- **Genomics Bioinformatics (2020-2023): Whole genome SNP/INDEL calling, transposable-element mapping, large scale RNA/miRNA/miRNA/lncRNA pipelines, multi-generation family inheritance mutation calling, possible biological meaning with literature search; single cell functional check** https://ax3301.github.io/RGC/rgc_2021.html**; dashboard development** https://ax3301.github.io/PDAC_2022
- **Web Scraping + Functional call: Toolkit for new medical physics students** https://ax3301.github.io/Medical_Physics_Toolkit_23/
- **NSF GRFP Proposal 2023: Large language models for advancements in biomedical research** https://ax3301.github.io/RGC/NSF_2023.pdf
- **NSF GRFP Proposal 2024: Precision livestock farming using engineering strategies in swine Barns** https://ax3301.github.io/RGC/NSF_2024.pdf
- **Water signal suppression in NMR imaging** https://github.com/ax3301/RGC/blob/main/NMR_2024.ipynb
- **Swine Vision: Imaging segmentation and time serious transformer neural network in swine** https://github.com/ax3301/Pig_Vision_2024
- **NSF New Grants Tracker and AI Agent**
  https://nsf-grant-tracker-2025.sequencingmaster.com/
  https://nsf-grant-tracker-2025.sequencingmaster.com/agent
- **California State University Agriculture Research Institution Student Fellowship Proposal 2025: Blockchain enabled agricultural data system with AI for swine and soybean production (Selected 10k)** https://ax3301.github.io/RGC/F_2025.pdf

Publication, Poster, Oral Presentation

- Chang H. (2021). A comprehensive list of undergraduate bioinformatics programs in the United States. https://doi.org/10.31219/osf.io/9g5wu
- Chang H., Q H. (Mar 2024). Visualization and algorithm for in vivo piSSelMQC signal processing in chemical shift imaging experiments: Recovering multi-quantum coherence transfer pathways. Presented at 2024 HSCI Purdue University Annual Research Retreat https://ax3301.github.io/RGC/HSCI_032024.pdf
- Chang H. (Apr 2024). The purpose of developing an online TG51 calculator for quick beam measurement: To provide a convenient and accessible tool for trainees. Oral presentation at: American Association of Physicists in Medicine ORVC 2024 Spring Meeting. https://ax3301.github.io/RGC/ORVC_2024.pdf

Technical Skills

Languages & Tools: Python, R, MATLAB, VHDL, SystemVerilog, HTM, Bash
Hardware & EDA: Microcontrollers, LiDAR/IR sensors, Virtuoso, Vivado
AI & ML: PyTorch, Transformers, OpenCV, Prompt Engineering
Bioinformatics: GATK, BWA, SAMtools, BCFtools, SnpEff, Pysam, Seurat, DESeq2
Data & Visualization: Pandas, Plotly, Dash, Matplotlib, ggplot2